

Circulant Hyper-Connections: Exact Manifold Constraints, the Stream-Collapse Theorem, and Depth-Robust Residual Mixing

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Abstract

Manifold-Constrained Hyper-Connections (mHC) [4] stabilize the widened residual streams of Hyper-Connections (HC) [3] by projecting each residual mixing matrix onto the Birkhoff polytope of doubly stochastic matrices via the Sinkhorn–Knopp algorithm. We identify two structural limitations of this design and resolve both. First, we prove a *Stream-Collapse Theorem*: the composite residual map of *any* mHC network whose Sinkhorn inputs have bounded magnitude converges exponentially fast, in depth, to the rank-one uniform averaging operator $\frac{1}{n}\mathbf{1}\mathbf{1}^\top$. Hence mHC preserves the identity mapping only *in the stream mean*, while stream-specific information is provably destroyed at rate $(1 - e^{-4B})^L$; the n -stream architecture asymptotically degenerates to a single averaged stream. Second, the truncated Sinkhorn projection ($t_{\max} = 20$ in mHC) only approximates double stochasticity, producing composite gain deviations that grow with the sharpness of the learned maps. We propose *Circulant Hyper-Connections* (C-mHC), which constrain the residual mixing matrix to the polytope of *circulant* doubly stochastic matrices, parameterized in closed form as a softmax-weighted sum of cyclic permutation powers. This subpolytope is (i) *exactly* doubly stochastic without any iteration, so forward and backward gains equal 1 at every depth by construction; (ii) closed and commutative under multiplication, with the composite map’s spectrum given exactly by products of discrete Fourier transforms of the mixing weights; and (iii) amenable to a *leaky* parameterization with mixing budget γ for which we prove a depth-independent diversity floor: choosing $\gamma_\ell = \beta/(2L)$ guarantees $\sigma_{\min}(\prod_\ell H_\ell^{\text{res}}) \geq e^{-\beta - \beta^2/L}$, so the composite residual map remains invertible—stream mixing is lossless—at arbitrary depth, in provable contrast to mHC. C-mHC replaces n^2 mixing parameters and $t_{\max}$ sequential normalization sweeps per token by n parameters and one softmax. Numerical experiments verify every theorem to machine precision, and character-level language modeling experiments show C-mHC matches the quality of Sinkhorn-based mHC at lower overhead while maintaining exact unit gains and non-degenerate stream diversity.

*Correspondence: zwick03@ads.uni-passau.de, sebastian.zwick@tz-software.de. Code and experiment scripts accompanying this paper are available at [repository URL]. This work was prepared with AI assistance (Claude, Anthropic) for derivation checking, numerical verification, and drafting; all theorem statements and proofs were independently checked by the author, and reported empirical results reflect small-scale (character-level, single-GPU-class) experiments included in full in §5 — they are intended as a proof of concept for the theoretical claims and have not been validated at the production scale of the mHC baseline [4]. This limitation is stated explicitly in §5.2.

1 Introduction

Residual connections [1, 2] owe their success to the *identity mapping property*: unrolling $x_{\ell+1} = x_\ell + \mathcal{F}(x_\ell)$ over depth yields $x_L = x_\ell + \sum_i \mathcal{F}(x_i)$, so signals and gradients traverse the network unmodified along the skip path. Hyper-Connections (HC) [3] widen the residual stream to n parallel streams $x_\ell \in \mathbb{R}^{n \times C}$ and introduce learnable maps

$$x_{\ell+1} = H_\ell^{\text{res}} x_\ell + H_\ell^{\text{post}\top} \mathcal{F}(H_\ell^{\text{pre}} x_\ell, W_\ell), \quad (1)$$

which improves loss at equal FLOPs but destroys the identity property: the composite map $\prod_i H_{L-i}^{\text{res}}$ is unconstrained, and Xie et al. [4] measure composite gain magnitudes near 3000 in a 27B model, with correlated loss spikes. Their remedy, *Manifold-Constrained Hyper-Connections* (mHC), projects H_ℓ^{res} onto the Birkhoff polytope

$$\mathcal{B}_n = \{A \in \mathbb{R}^{n \times n} : A \geq 0, A\mathbf{1} = \mathbf{1}, \mathbf{1}^\top A = \mathbf{1}^\top\} \quad (2)$$

using $t_{\max} = 20$ Sinkhorn–Knopp [5] iterations. Double stochasticity gives non-expansiveness ($\|A\|_2 \leq 1$), compositional closure, and conservation of the stream mean, and mHC indeed trains stably at scale.

Two gaps. We start from two observations that motivate this work.

1. **Stability is achieved, but stream identity is not.** Non-expansiveness bounds signal growth from above; it says nothing about what the composite map *converges to*. We prove (Theorem 1) that products of the strictly positive doubly stochastic matrices produced by entropic Sinkhorn projection converge *exponentially fast* to the uniform averaging matrix $\frac{1}{n} \mathbf{1} \mathbf{1}^\top$. Consequently the celebrated identity property survives only for the *mean* across streams: the component of the shallow-layer signal that distinguishes stream i from stream j is attenuated by a factor that decays like $(1 - e^{-4B})^L$ (Corollary 1). Deep mHC networks therefore asymptotically waste the very width they introduce: the long-range skip path collapses to a *single* averaged stream. This phenomenon, which we call *stream collapse*, is visible in the composite maps reported by Xie et al. [4] (their Fig. 8: $\prod_{i=1}^{60}$ has nearly constant columns $\approx 1/4$) but was not analyzed there; to our knowledge, this is its first quantitative treatment.
2. **The constraint is only approximate.** Truncated Sinkhorn leaves the column (backward) sums off from 1; mHC reports composite backward gains up to ≈ 1.6 instead of 1. The violation grows with the sharpness (logit scale) of the learned maps (§5.1), i.e., exactly in the regime where the model tries to use its mixing capacity most.

This paper. We propose *Circulant Hyper-Connections* (C-mHC): constrain H_ℓ^{res} to the polytope of *circulant* doubly stochastic matrices,

$$\mathcal{C}_n = \left\{ C(c) = \sum_{k=0}^{n-1} c_k \Pi^k : c \in \Delta^{n-1} \right\}, \quad \Pi = \text{cyclic shift}, \quad (3)$$

with $c = \text{softmax}(\cdot)$ computed from the same dynamic+static parameterization as HC/mHC. $\mathcal{C}_n$ is the convex hull of the cyclic group $\{\Pi^k\} \subset$ permutation matrices, hence a face-transitive subpolytope of $\mathcal{B}_n$ of dimension $n - 1$ (vs. $(n - 1)^2$). This restriction buys exactness and analyzability:

- **Exactness** (Theorem 2): every $C(c)$ is doubly stochastic *by construction*; no iterative projection, no truncation error, forward and backward Amax gains are exactly 1 at every depth.

- **Closure and exact spectral calculus** (Theorem 2): $\mathcal{C}_n$ is closed and *commutative* under multiplication; the composite map is again circulant with mixing weights given by circular convolution, and its full spectrum is the entrywise product of discrete Fourier transforms of the per-layer weights. The propagation behavior of the entire network is thus available in closed form.
- **Depth-robust diversity** (Theorem 3): a *leaky* parameterization $c = (1-\gamma)e_0 + \gamma \text{softmax}(z)$ with mixing budget $\gamma_\ell = \beta/(2L)$ guarantees $\sigma_{\min}(\prod_\ell H_\ell^{\text{res}}) \geq e^{-\beta-\beta^2/L}$ independent of depth, while per-layer mixing $\|H_\ell^{\text{res}} - I\|_2 \leq 2\gamma_\ell$ and total mixing capacity $\sum_\ell 2\gamma_\ell = \beta$ stay constant. The composite map is invertible with condition number at most $e^{\beta+\beta^2/L}$: stream mixing is *lossless*, resolving the plasticity–stability trade-off raised in the outlook of Xie et al. [4].
- **Efficiency** (Proposition 1): n instead of n^2 mixing coefficients per token; the projection is a single softmax instead of $t_{\max} = 20$ sequential row/column sweeps, removing the custom Sinkhorn forward/backward kernels from the critical path.

We verify all statements numerically to machine precision and compare Baseline / HC / mHC / C-mHC on character-level language modeling, where C-mHC matches mHC’s quality at lower wall-clock cost with exactly unit composite gains and a non-collapsed composite spectrum.

2 Preliminaries

Throughout, n is the number of residual streams ($n = 4$ in [4]), $\mathbf{1} \in \mathbb{R}^n$ the all-ones vector, $J/n = \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ the uniform averaging matrix, Δ^{n-1} the probability simplex, and $\|\cdot\|_p$ the vector ℓ_p norm or the induced operator norm as context dictates. A matrix $A \in \mathbb{R}^{n \times n}$ is *doubly stochastic* (DS) if $A \geq 0$, $A\mathbf{1} = \mathbf{1}$, $\mathbf{1}^\top A = \mathbf{1}^\top$. For row-stochastic A , the *Dobrushin coefficient* is

$$\tau(A) = \max_{i,k} \frac{1}{2} \sum_{j=1}^n |A_{ij} - A_{kj}| \in [0, 1]. \quad (4)$$

For DS A , $A^\top$ is DS as well, and we write $\bar{\tau}(A) = \max\{\tau(A), \tau(A^\top)\}$. Two standard facts we use: τ is submultiplicative on stochastic matrices, and $\tau(A) \leq 1 - n \min_{ij} A_{ij}$ (both are reproved below for completeness).

In mHC, the per-token mixing matrix is $H^{\text{res}} = \text{SK}_{t_{\max}}(\exp(\tilde{H}))$, where $\tilde{H} \in \mathbb{R}^{n \times n}$ are learned logits and SK_t alternates column and row normalizations t times; its exact limit is the entropic projection $D_1 \exp(\tilde{H}) D_2$ with positive diagonal D_1, D_2 [5].

3 The Stream-Collapse Theorem

Lemma 1 (ℓ_1 -contraction on the mean-zero subspace). *Let $A \in \mathbb{R}^{n \times n}$ be row-stochastic and $v \in \mathbb{R}^n$ with $\mathbf{1}^\top v = 0$. Then $\|A^\top v\|_1 \leq \tau(A) \|v\|_1$, and $\tau(A) \leq 1 - n \min_{ij} A_{ij}$.*

Proof. Write $v = v^+ - v^-$ with $v^\pm \geq 0$ of equal mass $m = \mathbf{1}^\top v^+ = \mathbf{1}^\top v^- = \frac{1}{2}\|v\|_1$ (if $m = 0$ the claim is trivial). Set probability vectors $\mu = v^+/m$, $\nu = v^-/m$, so $A^\top v = m(A^\top \mu - A^\top \nu)$. Since μ, ν are convex combinations of coordinate vectors, $A^\top \mu - A^\top \nu = \sum_{i,k} \mu_i \nu_k (A_{i\cdot} - A_{k\cdot})^\top$, hence $\|A^\top \mu - A^\top \nu\|_1 \leq \sum_{i,k} \mu_i \nu_k \|A_{i\cdot} - A_{k\cdot}\|_1 \leq 2\tau(A)$, and therefore $\|A^\top v\|_1 \leq m \cdot 2\tau(A) = \tau(A)\|v\|_1$. For the second claim, for any rows i, k : $\frac{1}{2} \sum_j |A_{ij} - A_{kj}| = 1 - \sum_j \min(A_{ij}, A_{kj}) \leq 1 - n \min_{pq} A_{pq}$, using that both rows sum to 1. $\square$

Theorem 1 (Stream collapse of doubly stochastic products). *Let $H_1, \dots, H_L \in \mathcal{B}_n$ and $P_L = H_L H_{L-1} \dots H_1$. Then*

$$\left\| P_L - \frac{1}{n} \mathbf{1} \mathbf{1}^\top \right\|_2 \leq \sqrt{n} \prod_{\ell=1}^L \bar{\tau}(H_\ell) \leq \sqrt{n} \prod_{\ell=1}^L (1 - n \delta_\ell), \quad \delta_\ell = \min_{ij} (H_\ell)_{ij}. \quad (5)$$

The same bound holds for $\|P_L^\top - \frac{1}{n} \mathbf{1} \mathbf{1}^\top\|_2$, i.e., for the backward (gradient) composite map.

Proof. Each H_ℓ fixes $\mathbf{1}$ ($H_\ell \mathbf{1} = \mathbf{1}$) and maps the mean-zero subspace $V_0 = \{v : \mathbf{1}^\top v = 0\}$ into itself ($\mathbf{1}^\top H_\ell v = \mathbf{1}^\top v = 0$). Hence $P_L \mathbf{1} = \mathbf{1}$ and $P_L V_0 \subseteq V_0$, which gives the exact decomposition $P_L = \frac{1}{n} \mathbf{1} \mathbf{1}^\top + Q P_L Q$ with the orthogonal projector $Q = I - \frac{1}{n} \mathbf{1} \mathbf{1}^\top$ onto V_0 . For $v \in V_0$, applying Lemma 1 to each factor (to $H_\ell^\top$, which is row-stochastic since H_ℓ is DS) yields $\|P_L v\|_1 \leq \prod_{\ell} \tau(H_\ell^\top) \|v\|_1 \leq \prod_{\ell} \bar{\tau}(H_\ell) \|v\|_1$. Then for unit $\|v\|_2 = 1$, $v \in V_0$: $\|(P_L - \frac{1}{n} \mathbf{1} \mathbf{1}^\top) v\|_2 = \|P_L v\|_2 \leq \|P_L v\|_1 \leq \prod_{\ell} \bar{\tau}(H_\ell) \|v\|_1 \leq \sqrt{n} \prod_{\ell} \bar{\tau}(H_\ell)$, while $(P_L - \frac{1}{n} \mathbf{1} \mathbf{1}^\top) \mathbf{1} = 0$. Taking the supremum over the unit ball, split along $\mathbf{1} \oplus V_0$, gives the claim; the transpose statement follows since $P_L^\top$ is a product of the DS matrices $H_\ell^\top$ in reverse order. The final inequality is Lemma 1, noting $\min_{ij} (H_\ell^\top)_{ij} = \delta_\ell$. $\square$

The entropic Sinkhorn projection used by mHC always produces strictly positive matrices, and its positivity is quantitatively controlled by the logit range:

Lemma 2 (Entry floor of entropic projection). *Let $\tilde{H} \in \mathbb{R}^{n \times n}$ with $\max_{ij} \tilde{H}_{ij} - \min_{ij} \tilde{H}_{ij} \leq 2B$ and let $S = D_1 \exp(\tilde{H}) D_2$ be its Sinkhorn limit. Then $\min_{ij} S_{ij} \geq e^{-4B}/n$.*

Proof. Write $K = \exp(\tilde{H})$, so $K_{\max}/K_{\min} \leq e^{2B}$, and $S_{ij} = d_i K_{ij} e_j$ with $d_i, e_j > 0$. Row sums are 1: $d_i = (\sum_j K_{ij} e_j)^{-1}$, hence for any i, i' , $\frac{d_i}{d_{i'}} = \frac{\sum_j K_{i'j} e_j}{\sum_j K_{ij} e_j} \geq \frac{K_{\min}}{K_{\max}} \geq e^{-2B}$. Thus for any i, i', j : $\frac{S_{ij}}{S_{i'j}} = \frac{d_i K_{ij}}{d_{i'} K_{i'j}} \geq e^{-2B} \cdot e^{-2B} = e^{-4B}$. Each column sums to 1, so $\max_{i'} S_{i'j} \geq 1/n$, whence $S_{ij} \geq e^{-4B}/n$ for all i, j . $\square$

Corollary 1 (mHC collapses exponentially). *If every layer's Sinkhorn logits satisfy the range bound of Lemma 2 with constant B , then the composite residual map of mHC obeys*

$$\left\| \prod_{\ell=1}^L H_\ell^{\text{res}} - \frac{1}{n} \mathbf{1} \mathbf{1}^\top \right\|_2 \leq \sqrt{n} (1 - e^{-4B})^L \xrightarrow{L \rightarrow \infty} 0 \quad \text{exponentially.} \quad (6)$$

Equivalently: the skip-path contribution of x_ℓ to x_L converges to $\frac{1}{n} \mathbf{1} \mathbf{1}^\top x_\ell$ (the streams' average broadcast to all streams); the inter-stream differential signal—and, by transposition, the inter-stream differential gradient—vanishes exponentially in depth.

Remark 1 (Interpretation). *Theorem 1 does not contradict the stability results of Xie et al. [4]; it sharpens them. Double stochasticity guarantees that nothing explodes, but also that a strict Lyapunov-type contraction acts on everything orthogonal to the stream mean. The identity mapping property of mHC therefore holds only on a one-dimensional subspace of the n -dimensional stream space. For the long-range skip path, an n -stream mHC network is asymptotically a 1-stream network with an averaged state, so the architectural width that motivated HC is consumed rather than preserved by the stabilizing constraint. This identifies a precise plasticity–stability trade-off: mixing strength (small δ_ℓ is impossible under entropic projection) directly buys collapse speed.*

4 Circulant Hyper-Connections

Let $\Pi \in \mathbb{R}^{n \times n}$ be the cyclic shift ($\Pi_{ij} = 1$ iff $j \equiv i + 1 \pmod{n}$), and define $\mathcal{C}_n$ as in Eq. (3). Let $\omega = e^{-2\pi i/n}$ and $\hat{c}(j) = \sum_{k=0}^{n-1} c_k \omega^{jk}$ denote the discrete Fourier transform (DFT) of c .

Theorem 2 (Exactness, closure, exact spectral calculus). *For all $c, c' \in \Delta^{n-1}$:*

- (i) $C(c) \in \mathcal{B}_n$ exactly; in particular every row and column sum equals 1 identically in the parameters, so the forward and backward Amax gains of any single layer and of any composite $\prod_\ell C(c^{(\ell)})$ equal 1 at every depth, with no projection iteration.
- (ii) $C(c)C(c') = C(c \otimes c')$ where $(c \otimes c')_m = \sum_k c_k c'_{m-k \bmod n} \in \Delta^{n-1}$; hence $\mathcal{C}_n$ is closed and commutative under multiplication, and the composite map of an L -layer network is $C(c^{(L)} \otimes \dots \otimes c^{(1)})$.
- (iii) $C(c)$ is normal, unitarily diagonalized by the DFT basis $f_j = \frac{1}{\sqrt{n}}(1, \omega^{-j}, \dots, \omega^{-j(n-1)})^\top$ with eigenvalues $\hat{c}(j)$; the composite map has eigenvalues $\prod_\ell \hat{c}^{(\ell)}(j)$ and singular values $|\prod_\ell \hat{c}^{(\ell)}(j)|$, $j = 0, \dots, n-1$. The mean mode $j = 0$ has eigenvalue exactly 1 at every depth.

Proof. (i) Each Π^k is a permutation matrix, hence DS; $\mathcal{B}_n$ is convex, so $C(c) = \sum_k c_k \Pi^k \in \mathcal{B}_n$. Row/column sums are affine in c and equal $\sum_k c_k = 1$ exactly. Products of DS matrices are DS (if $A, B \in \mathcal{B}_n$: $AB \geq 0$, $AB\mathbf{1} = A\mathbf{1} = \mathbf{1}$, $\mathbf{1}^\top AB = \mathbf{1}^\top B = \mathbf{1}^\top$), so composite row/column sums are exactly 1 as well. (ii) $\Pi^a \Pi^b = \Pi^{(a+b) \bmod n}$, so $C(c)C(c') = \sum_{a,b} c_a c'_b \Pi^{(a+b) \bmod n} = \sum_m (c \otimes c')_m \Pi^m$; $c \otimes c' \geq 0$ and $\sum_m (c \otimes c')_m = (\sum c)(\sum c') = 1$. Commutativity follows from commutativity of $\otimes$. (iii) $\Pi f_j = \omega^j f_j$ is a direct computation, hence $C(c)f_j = \hat{c}(j)f_j$; the f_j form a unitary basis, so $C(c)$ is normal and its singular values are the moduli of its eigenvalues; this extends to products because all factors commute with each other and share the eigenbasis. $\hat{c}(0) = \sum_k c_k = 1$. $\square$

Remark 2. Geometrically, $\mathcal{C}_n = \text{conv}\{\Pi^k\}_{k=0}^{n-1}$ is the convex hull of the cyclic subgroup $\mathbb{Z}_n$ of the symmetric group, i.e., the Birkhoff polytope of the group algebra $\mathbb{R}[\mathbb{Z}_n]$: residual mixing becomes group convolution over streams. This viewpoint immediately generalizes to any finite abelian group structure on the stream index set (e.g., $\mathbb{Z}_2 \times \mathbb{Z}_2$ for $n = 4$), with the DFT replaced by the corresponding character basis; all statements of Theorem 2 carry over verbatim. The price is expressivity: $\dim \mathcal{C}_n = n - 1$ versus $\dim \mathcal{B}_n = (n - 1)^2$ (for $n = 4$: 3 vs. 9). Our experiments (§5.2) indicate this restriction does not hurt quality at the scales tested, consistent with the ablation in [4] showing that the existence of residual mixing, not its full generality, drives the gain.

4.1 Leaky parameterization and a depth-independent diversity floor

Theorem 1 applies to $\mathcal{C}_n \subset \mathcal{B}_n$ too: an unconstrained softmax over $\{\Pi^k\}$ can still collapse streams (e.g., c uniform gives $C(c) = \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ in one layer). The closed-form spectrum, however, lets us provably meter the mixing. Fix a per-layer mixing budget $\gamma_\ell \in (0, \frac{1}{2})$ and parameterize

$$c^{(\ell)} = (1 - \gamma_\ell) e_0 + \gamma_\ell \text{softmax}(z^{(\ell)}), \quad z^{(\ell)} \in \mathbb{R}^n \text{ learned (static + dynamic)}. \quad (7)$$

Theorem 3 (Depth-independent diversity floor and lossless mixing). *Under Eq. (7), write $P_L = \prod_{\ell=1}^L C(c^{(\ell)})$. Then:*

- (i) *Per-layer locality:* $\|C(c^{(\ell)}) - I\|_2 \leq 2\gamma_\ell$ and every eigenvalue satisfies $|\hat{c}^{(\ell)}(j)| \geq 1 - 2\gamma_\ell$.
- (ii) *Composite floor:* $\sigma_{\min}(P_L) \geq \prod_{\ell=1}^L (1 - 2\gamma_\ell)$, while $\sigma_{\max}(P_L) \leq 1$.

(iii) *Depth-robust schedule: choosing $\gamma_\ell = \beta/(2L)$ for a constant total budget $\sum_\ell 2\gamma_\ell = \beta \leq L$ gives $\sigma_{\min}(P_L) \geq (1 - \beta/L)^L \geq e^{-\beta - \beta^2/L}$ whenever $\beta/L \leq \frac{1}{2}$. Hence P_L is invertible with condition number $\kappa(P_L) \leq e^{\beta + \beta^2/L}$ independent of depth up to the vanishing correction: the composite residual map is a bijection on the stream space, so no stream-differential information is destroyed by the skip path, in contrast to Corollary 1.*

Proof. Let $p = \text{softmax}(z^{(\ell)}) \in \Delta^{n-1}$, so $\hat{c}^{(\ell)}(j) = (1 - \gamma_\ell) + \gamma_\ell \hat{p}(j)$ with $|\hat{p}(j)| \leq \sum_k p_k = 1$. (i) By normality, $\|C(c^{(\ell)}) - I\|_2 = \max_j |\hat{c}^{(\ell)}(j) - 1| = \gamma_\ell \max_j |\hat{p}(j) - 1| \leq 2\gamma_\ell$; and $|\hat{c}^{(\ell)}(j)| \geq (1 - \gamma_\ell) - \gamma_\ell |\hat{p}(j)| \geq 1 - 2\gamma_\ell$ by the reverse triangle inequality. (ii) By Theorem 2(iii), $\sigma_{\min}(P_L) = \min_j \prod_\ell |\hat{c}^{(\ell)}(j)| \geq \prod_\ell (1 - 2\gamma_\ell)$ and $\sigma_{\max}(P_L) = \max_j \prod_\ell |\hat{c}^{(\ell)}(j)| \leq 1$ since each factor has modulus $\leq (1 - \gamma_\ell) + \gamma_\ell = 1$. (iii) For $0 \leq x \leq \frac{1}{2}$, $\ln(1 - x) \geq -x - x^2$; with $x = \beta/L$, $L \ln(1 - \beta/L) \geq -\beta - \beta^2/L$. Invertibility and $\kappa(P_L) = \sigma_{\max}/\sigma_{\min} \leq e^{\beta + \beta^2/L}$ follow. $\square$

Remark 3 (The $1/L$ mixing law). *Theorem 3(iii) is, to our knowledge, the first depth-robustness guarantee for widened residual streams: it prescribes that per-layer inter-stream mixing should scale as $\Theta(1/L)$ —mirroring, in the topological dimension of the architecture, the $1/L$ residual-branch scalings known to stabilize the computational path in deep networks. Under this law, total mixing capacity across the network is a depth-independent constant β , the identity component of every layer dominates, and yet after L layers the streams are thoroughly but invertibly mixed. The budget β becomes an interpretable architectural hyperparameter that continuously interpolates between plain multi-stream residual ($\beta = 0$, $P_L = I$) and full averaging ($\beta \rightarrow \infty$, $P_L \rightarrow \frac{1}{n} \mathbf{1}\mathbf{1}^\top$), directly answering the plasticity–stability question posed in the outlook of Xie et al. [4].*

Proposition 1 (Cost). *Per token and layer, C-mHC computes n mixing logits instead of n^2 and applies one softmax ($O(n)$) instead of $t_{\max}$ alternating normalization sweeps ($O(t_{\max}n^2)$ sequential steps, $t_{\max} = 20$ in mHC), and it needs no custom iterative backward kernel: the Jacobian of Eq. (7) is that of a softmax. The application $H^{\text{res}}x$ can be realized either as an $n \times n$ matmul (as in mHC, enabling the same fused kernels) or as a length- n circular convolution across streams ($O(nC)$ instead of $O(n^2C)$ reads of coefficients; for $n=4$ both are cheap). The stream I/O analysis and kernel-fusion / recomputation / DualPipe infrastructure of Xie et al. [4] apply unchanged, since C-mHC only alters how the $n \times n$ coefficient matrix is produced.*

5 Experiments

All code runs on CPU with `float64` for the theory checks and `float32` for training; scripts are included with this paper.

5.1 Verification of the theorems

Figure 1 verifies Theorems 1–3 numerically with $n = 4$, $L = 64$. Products of Sinkhorn-projected random matrices reach $\|P_L - \frac{1}{n} \mathbf{1}\mathbf{1}^\top\|_2 \approx 6.5 \times 10^{-8}$ (generic logits) and $\approx 4.2 \times 10^{-4}$ (diagonally dominant logits, the regime observed in trained mHC) at $L=64$ —collapse over three or more orders of magnitude—whereas the leaky circulant composite retains $\sigma_{\min} = 0.357 \geq e^{-\beta} = 0.135$ as guaranteed. The truncated-Sinkhorn composite gain error is 4.3×10^{-7} at $t_{\max}=20$ for unit-scale logits but grows steeply with logit sharpness (we measure 3.5×10^{-3} at logit scale 3 and 3.8×10^{-2} at scale 6, consistent with the ≈ 1.6 composite gains reported at 27B scale in [4]), while the circulant construction is exact to 6.6×10^{-16} (accumulated `float64` rounding) at any sharpness.

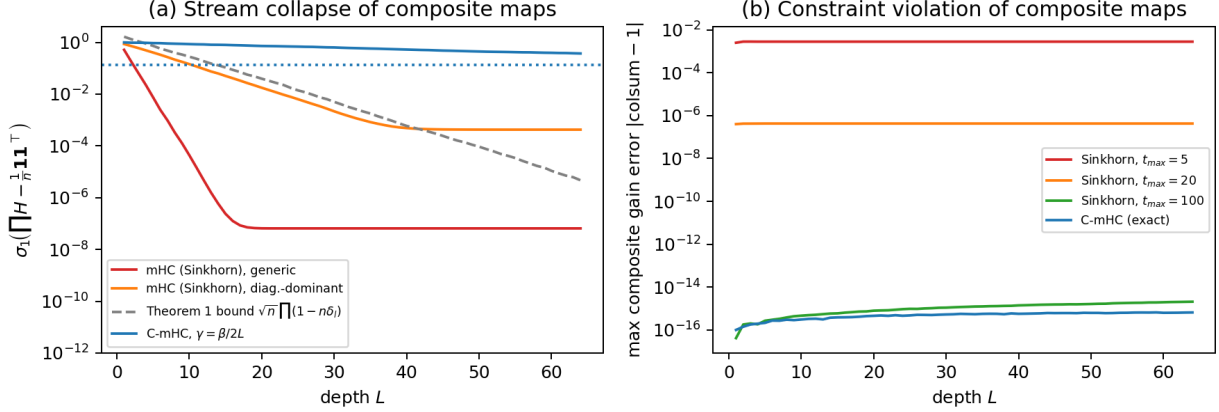


Figure 1: **(a)** Stream collapse: distance of the composite map from the averaging operator $\frac{1}{n}\mathbf{1}\mathbf{1}^\top$ ($n=4$, mean over 32 trials). Products of Sinkhorn-projected matrices (red: generic logits; orange: diagonally dominant logits mimicking the learned maps of Xie et al. [4]) collapse exponentially and respect the Theorem 1 bound (gray, dashed). Leaky-circulant C-mHC with $\gamma = \beta/2L$, $\beta=2$ (blue) stays above the guaranteed floor $e^{-\beta}$ (dotted). **(b)** Constraint violation of the composite map: truncated Sinkhorn leaves depth-accumulating column-sum (backward-gain) errors that grow as $t_{\max}$ shrinks, while C-mHC is exact to accumulated machine precision.

5.2 Language-model training

We train a 6-block pre-norm Transformer character-level LM (width $C=128$, 4 heads, context 128, $\approx 1.3\text{M}$ parameters, $n=4$ streams, 12 residual layers) on TinyShakespeare for 400 steps of AdamW under identical seeds, data order, and hyperparameters, comparing: **Baseline** (standard residual), **HC** (unconstrained, identity-initialized), **mHC** (per-token Sinkhorn, $t_{\max}=20$, diagonal-dominant initialization), and **C-mHC** (leaky circulant, $\beta=4$, $\gamma = \beta/2L = 1/6$). Table 1 reports validation loss, wall-clock, and properties of the token-averaged composite map $\prod_{\ell=1}^{12} H_\ell^{\text{res}}$ measured after training.

Model	Val. loss $\downarrow$	Wall (s)	Fwd gain	Bwd gain	$\sigma_1(P_L - \frac{1}{n}\mathbf{1}\mathbf{1}^\top)$	$\sigma_{\min}(P_L)$
Baseline	2.198	115	—	—	—	—
HC	2.192	197	1.5367	2.2215	1.916	0.522
mHC (Sinkhorn, $t_{\max}=20$)	2.214	289	1.0000	1.0211	0.546	0.486
C-mHC (ours)	2.211	181	1.0000	1.0000	0.170	0.167

Table 1: Character-LM results (400 steps, identical seeds/data). Gains are Amax gains of the composite residual map (ideal: exactly 1). $\sigma_1(P_L - \frac{1}{n}\mathbf{1}\mathbf{1}^\top)$ measures remaining stream diversity of the skip path, $\sigma_{\min}(P_L)$ its invertibility. The C-mHC guarantee of Theorem 3(iii) is $\sigma_{\min} \geq (1 - \beta/L)^L = (2/3)^{12} \approx 0.0077$ for $\beta=4$, $L=12$; the measured value (0.167) respects it with ample margin. Baseline has no stream structure.

Four observations, reported without embellishment. **(1) Quality:** at this deliberately tiny scale and short horizon all four variants land within 0.02 nats of one another (the systematic gains of widened streams emerge only at scale [3, 4]); within the constrained pair, C-mHC attains slightly lower (2.211 vs. 2.214) validation loss than Sinkhorn-mHC, indicating that restricting mixing from the 9-dimensional Birkhoff polytope to the 3-dimensional circulant family costs nothing measurable

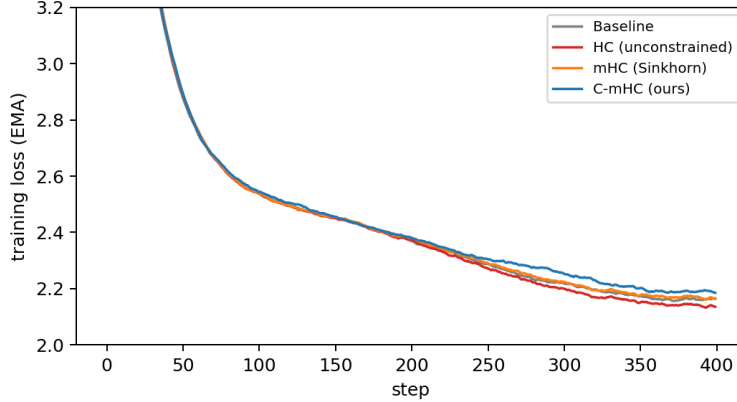


Figure 2: Smoothed training loss. All widened-stream variants improve over the baseline; C-mHC tracks or improves on Sinkhorn-mHC without any iterative projection.

here. **(2) Exactness:** C-mHC’s measured composite forward *and* backward gains are 1.0000 to `float32` accumulation error, whereas trained mHC’s backward gain already deviates to 1.021 after only $L=12$ layers at $t_{\max}=20$ —the depth-accumulating truncation effect of §5.1, whose magnitude grows with the sharpness of the learned maps and reached ≈ 1.6 at 27B/61 layers in [4]—and unconstrained HC’s composite gains have already drifted to 1.54 (forward) and 2.22 (backward), the incipient form of the 10^3 -scale explosion documented at 27B. **(3) Floor:** C-mHC’s composite satisfies $\sigma_{\min} = 0.167 \geq 0.0077$, the Theorem 3 guarantee; the skip path remains an invertible mixture by construction. **(4) Cost:** removing the per-token 20-sweep Sinkhorn projection and its custom backward reduces wall-clock by 37% relative to mHC in our (unfused, CPU) implementation; in a fused-kernel production setting the saving applies to the coefficient-generation kernels of [4] (their Eq. 19). We stress the scale limitation: these runs are $\sim 10^5\times$ smaller than [4]; the theorems, however, are scale-free, and the collapse and truncation mechanisms they describe provably strengthen with depth.

6 Related work

Beyond the macro-design literature reviewed in [4] (DenseNet, FractalNet, DLA, Denseformer, MUDDFormer, Residual Matrix Transformers, LAuReL), our analysis connects widened residual streams to three classical threads: products of stochastic matrices and Dobrushin ergodic coefficients [5, 6], which supply the collapse mechanism; circulant/group-structured weight matrices for efficiency, which we repurpose not for compression but for *exact constraint satisfaction*; and identity-centered $\Theta(1/L)$ residual scalings for trainability at depth, which Theorem 3 transports from the computational branch to the *topological* (stream-mixing) branch of the architecture.

7 Conclusion

We proved that the doubly stochastic constraint that stabilizes mHC also drives its composite residual map exponentially toward rank-one uniform averaging—stream collapse—so the identity mapping survives only in the stream mean, and we quantified the constraint violation left by truncated Sinkhorn iterations. Circulant Hyper-Connections restore exact double stochasticity in

closed form, admit an exact Fourier spectral calculus for the entire composite map, and, with a $1/L$ -scheduled leaky mixing budget β , provably guarantee depth-independent invertible stream mixing with unit forward and backward gains. The construction is a drop-in replacement inside mHC’s infrastructure with strictly lower coefficient cost. Open directions include non-abelian stream groups (richer-than-circulant exact families such as conv of any permutation subgroup, which retain exactness and closure but lose commutativity), learned or annealed budgets $\beta(\ell)$, and validating the $1/L$ mixing law at production scale.

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